Multiple Sequence Alignments

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Recap

- Global Alignment: Maximize overall matches
- Local Alignment: Maximize concentrated matches
- Semi-global: Disregard start and/or end gaps
- Database search: find matches with heuristics

Multiple Sequence Alignment

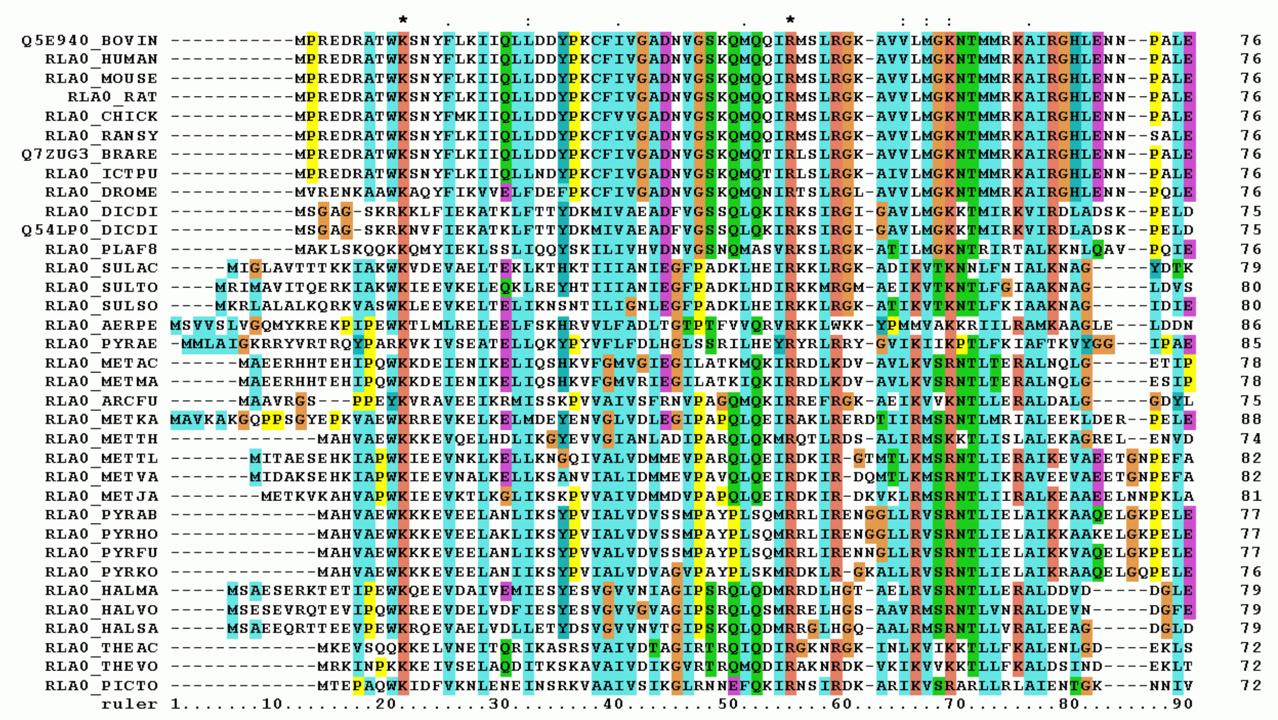
- Pairwise alignment: infer function from sequence similarity.
- Multiple alignments (MSA):
  - given that a group of protein sequences have similar functions, we would like to know which sequence features are essential for this function
- MSA can elucidate similarities that could not be found with pairwise comparisons
- Essential sequence motifs should be conserved, i.e. be similar, between homologous sequences from diverse species.
- The easiest way to discover conservation is to align many sequences encoding the function of interest and to look for invariant regions.

Multiple Sequence Alignment

- An alignment of more than two sequences simultaneously.
- Given a set of sequences $X_1, \dots, X_n$ with $n > 2$:
  - find a set $X'_1, \dots, X'_n$ where X'_i augments X_i with gaps –
  - that maximizes the alignment score $S(X'_1, \dots, X'_n)$
  - the score S depends on gaps and matches per column
  - no column can be all –
- MSAs can be global or local, as it happens with their pairwise counterparts. Here we deal with global MSA.
- The concept of local MSA is strongly related to motif discovery that will be addressed later.

Multiple Sequence Alignment

- A multiple sequence alignment, showing the first 90 positions of sequences of a ribosomal protein from several organisms.



Multiple Sequence Alignment

- To find the best alignment we need:
  - Some way to score each possible MSA
  - Some way to search possible MSAs to find the best

Entropy Score

- Shannon binary entropy function for a random variable:

$$H(X) = -\sum_{i=1}^{A} P(x_i) \times \log_2 P(x_i)$$

  - This function measures how unpredictable the values are, with a maximum entropy corresponding to a uniform distribution.

- Score for the column i of MSA m is:

$$S(m_i) = -\sum_{a=1}^{A} c_{ia} \times \log p_{ia}$$

  - where c_{ia} is the number of occurrences of symbol a in column i and p_{ia} is the probability of occurrence of symbol a in column i, which is simply the fraction of times a occurs in column i.

- The best MSA is the one that **minimizes** entropy.

Entropy Score

- Compute the entropy score for the fourth position, containing one T and three gaps:

```
1  A C A T T A A C G A T
2  A G - - T A A C G A -
3  A G A - T C A A G - T
4  A C - - T A A - - - T
```

$$S(m_i) = -\sum_{a=1}^{A} c_{ia} \times \log p_{ia}$$

Sum-of-pairs (SP) Score

- Use a substitution matrix to score each pair.
- The SP score of column i is the sum of the substitution scores for all pairs:

$$SP(m_i) = \sum_{j=1}^{N-1} \sum_{k=j+1}^{N} S(m_i^j, m_i^k)$$

  - where $S(m_i^j, m_i^k)$ is the substitution score for the two residues on rows j and k of column i.
  - To account for gaps, the substitution matrix also includes scores for pairing each residue type with a gap.
- The best MSA is the one that maximizes SP.

Sum-of-pairs (SP) Score

- Compute SP score for the second column using a simple substitution matrix that adds 1 for each identity and -1 if the symbols are different:

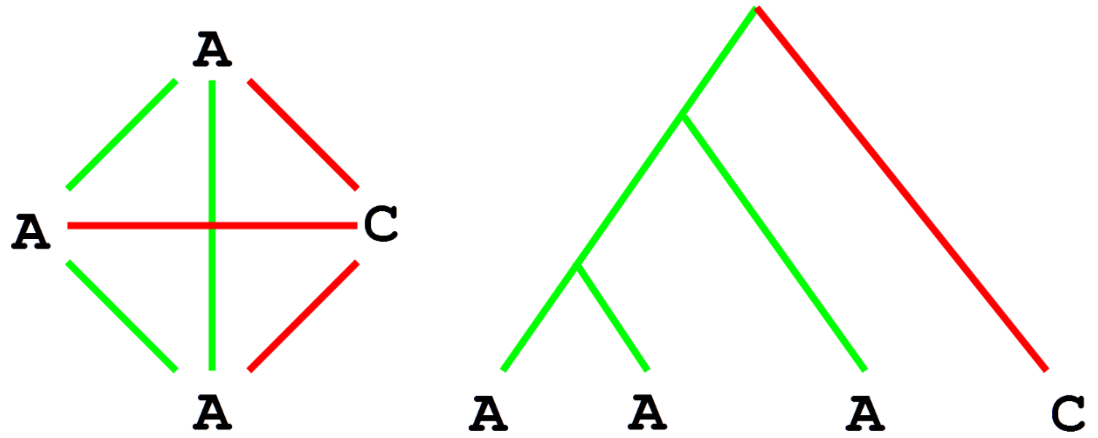
```
1  A C A T T A A C G A T
2  A G - - T A A C G A -
3  A G A - T C A A G - T
4  A C - - T A A - - - T
     *
```

$$SP(m_i) = \sum_{j=1}^{N-1} \sum_{k=j+1}^{N} S(m_i^j, m_i^k)$$

Sum-of-pairs (SP) Score

- Often does not reflect the evolution process and overestimates mutations.
- Example:

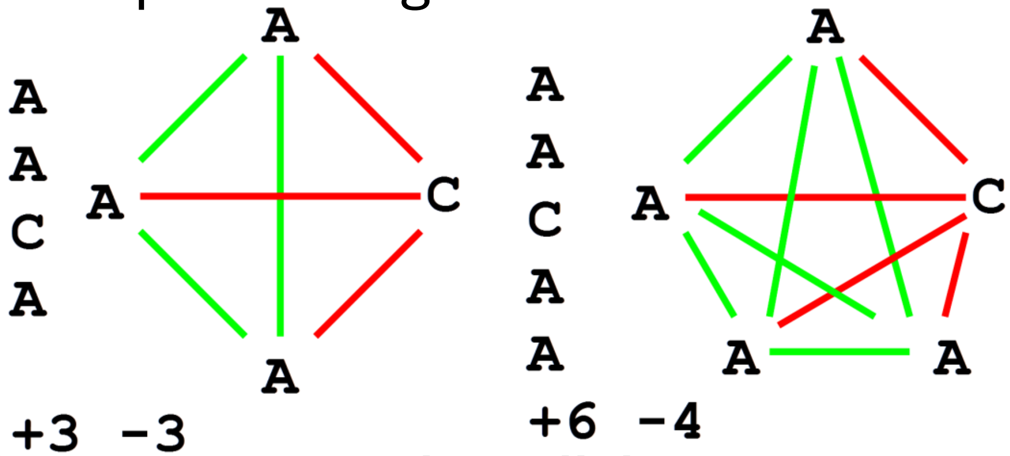
A
A
C
A



  - Although the C residue could be explained with a single mutation in the evolutionary tree (right panel), the SP score counts it 3 times when comparing to each of the A residues.

Sum-of-pairs (SP) Score

- Another problem: counter-intuitive result of giving a smaller relative weight to a disagreement the more sequences agree.
- Example:



  - Left: Disagreeing C residue counts **as much as all the agreements**
  - Right: we increase the conserved residues with more sequences, the **relative weight of the disagreement in this column decreases**.
  - This is counter-intuitive because **the more conserved a region is the greater the significance of a mismatch in that region**.

Dynamic Programming for MSA

- Dynamic Programming for pairwise sequence alignment: solves those optimization problems, assuring optimal solutions.
- Can we generalize these algorithms for N sequences?
- Aligning 2 sequences:
  - use a 2D matrix, 3 Neighbors for optimal path

$$F_{i,j} = max \begin{cases} F_{i-1,j-1} + S(A_i, B_j) \\ F_{i,j-1} + S(A_i, -) \\ F_{i-1,j} + S(-, B_j) \end{cases}$$

Dynamic Programming for MSA (BA5M)

- Can we generalize these algorithms for 3 sequences?
- Aligning 3 sequences:
  - need a 3D matrix, 7 Neighbors for optimal path

$$F_{i,j,k} = max \begin{cases} F_{i-1,j-1,k-1} + S(A_i, B_j, C_k) \\ F_{i,j-1,k-1} + S(-, B_j, C_k) \\ F_{i-1,j,k-1} + S(A_i, -, C_k) \\ F_{i-1,j-1,k} + S(A_i, B_j, -) \\ F_{i,j,k-1} + S(-, -, C_j) \\ F_{i,j-1,k} + S(-, B_j, -) \\ F_{i-1,j,k} + S(A_i, -, -) \end{cases}$$

Dynamic Programming for MSA

- Can we generalize these algorithms for N sequences?
- Aligning N sequences:
  - need a N-dimension matrix, L^N cells
  - 2^N-1 Neighbours per cell
  - Time complexity is $O(L^N 2^N)$
  - Aligning 10 sequences of length 100 takes 10^{18} times longer than aligning 2 sequences of length 100
- Multiple Sequence Alignment cannot be solved with the same algorithms used to align 2 sequences.

Heuristic Multiple Alignment

- Compute only pairwise alignments.
- Comparison order corresponding to the evolutionary relationships of the sequences to be aligned.
- Unable to assure optimal solutions, but have an acceptable running time complexity.

Heuristic Multiple Alignment

- Heuristic algorithms for MSA can be classified as:
  - Progressive (hierarchical) – sequence by sequence
  - Iterative – step by step
  - Hybrid – combine different strategies, and use complementary information (e.g. protein structural information, libraries of good local alignments).

Progressive (Hierarchical) Algorithms

- Create an **initial alignment** of the two most related sequences, adding, in subsequent iterations, increasingly distant sequences, until the final alignment with all sequences is attained.
- General idea: build a **guide tree** from the set of all possible pairwise alignments.
  - This tree groups closer together sequences that are more alike and places at a greater distance those more different.
  - Start by aligning the closest sequences according to tree and then group the closest groups, hierarchically through the tree

Progressive (Hierarchical) Algorithms

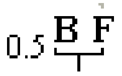
- CLUSTAL: widely used progressive MSA program that has given rise to a successful family of algorithms, namely CLUSTALW and Clustal Omega.
- In the original version, it used a fast pairwise alignment algorithm to compute all pairwise alignments and then computed a guide tree with the **Unweighted Pair Group Method with Arithmetic Mean** algorithm (UPGMA).
- UPGMA algorithm consists of:
  - picking the two closest groups,
  - joining the two as a single group and then
  - computing its distance to all other groups as the average of the distances of each element in the group to all outside groups.

Progressive (Hierarchical) Algorithms

- Example

	A	B	C	D	E	F	G
A							
B	19.00						
C	27.00	31.00					
D	8.00	18.00	26.00				
E	33.00	36.00	41.00	31.00			
F	18.00	**1.00**	32.00	17.00	35.00		
G	13.00	13.00	29.00	14.00	28.00	12.00	

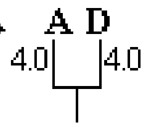
JOIN B, F →



	A	BF	C	D	E	G
A						
BF	18.50					
C	27.00	31.50				
D	8.00	17.50	26.00			
E	33.00	35.50	41.00	31.00		
G	13.00	12.50	29.00	14.00	28.00	

	A	BF	C	D	E	G
A						
BF	18.50					
C	27.00	31.50				
D	**8.00**	17.50	26.00			
E	33.00	35.50	41.00	31.00		
G	13.00	12.50	29.00	14.00	28.00	

JOIN A, D →



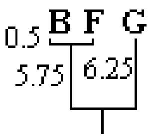
	AD	BF	C	E	G
AD					
BF	18.00				
C	26.50	31.50			
E	32.00	35.50	41.00		
G	13.50	12.50	29.00	28.00	

Progressive (Hierarchical) Algorithms

- Example

	AD	BF	C	E	G
AD					
BF	18.00				
C	26.50	31.50			
E	32.00	35.50	41.00		
G	13.50	**12.50**	29.00	28.00	

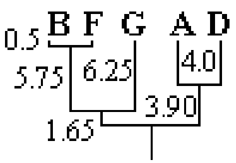
JOIN BF, G →



	AD	BFG	C	E
AD				
BFG	15.80			
C	26.50	30.30		
E	32.00	31.80	41.00	

	AD	BFG	C	E
AD				
BFG	**15.80**			
C	26.50	30.30		
E	32.00	31.80	41.00	

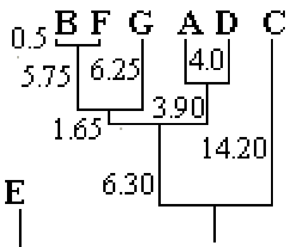
JOIN AD, BFG →



	ADBFG	C	E
ADBFG			
C	28.40		
E	31.90	41.00	

	ADBFG	C	E
ADBFG			
C	**28.40**		
E	31.90	41.00	

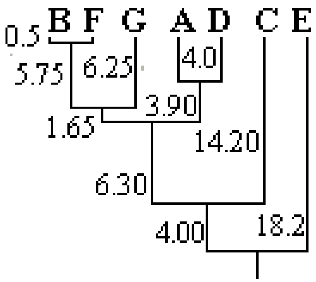
JOIN ADBFG, C →



	ADBFGC	E
ADBFGC		
E	36.40	

	ADBFGC	E
ADBFGC		
E	**36.40**	

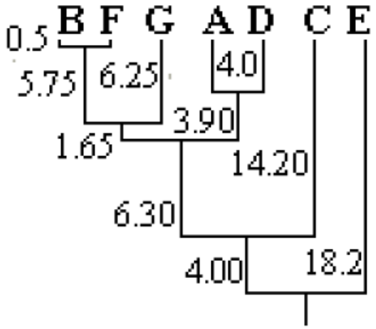
JOIN ADBFGC, E →



Finished

Progressive (Hierarchical) Algorithms

- The selection of the two first sequences is of foremost importance in these methods: the core of the alignment.
- These sequences are the ones with higher similarity, with their common ancestor nearer the leaves of the tree.
- The order of the remaining sequences to consider by the algorithm will be defined by the guide tree.
- Example:



Progressive (Hierarchical) Algorithms

- How further sequences are added to an existing alignment?
  - Create, from the existing alignment, a **summary of the content of each column**
  - In each iteration, align a sequence with a previous alignment, represented as a **consensus**
- Another alternative is to take into account the frequency of each symbol at each position of the current alignment
  - algorithm similar to the ones used in pairwise alignment, adjusted to align a sequence to a **profile –** we will talk about this in a few weeks

Progressive (Hierarchical) Algorithms

CLUSTAL-W

- Mitigated the problems with the sum-of-pairs
- Added the option to replace the UPGMA clustering method with the Neighbour Joining (NJ) algorithm.
  - UPGMA can be much faster
  - NJ can give better results

Progressive (Hierarchical) Algorithms

CLUSTAL-OMEGA

- Uses Hidden Markov Models (HMM) to create sequence profiles in the alignment.
- Uses a faster clustering algorithm for the guide tree.
    - allows the algorithm to be usable with a larger number of sequences

Iterative Algorithms for MSA

- Progressive alignment: errors that occur in the early stages cannot be corrected and may have a large impact on the resulting MSA.
- Iterative alignment algorithms try to remedy this problem by going back and forth in the hierarchical alignment to try to find and fix these mistakes.

Iterative Algorithms for MSA

- Start by generating an alignment using a given method, which may be a progressive alignment or any other method.
- try to improve this alignment by **making selected changes** and **evaluating the effect of those changes** over the objective function.
- repeatedly realign sub-groups of sequences, or sub-groups of columns, by **changing the position of gaps or adding/removing gaps**.

Hybrid Algorithms for MSA

- Combine progressive alignments based on profiles, with techniques from iterative methods.

MUSCLE

- MUltiple Sequence Comparison by Log-Expectation
- Is one of the more efficient methods currently available.
- Works by refining the guide tree to improve the resulting alignment.

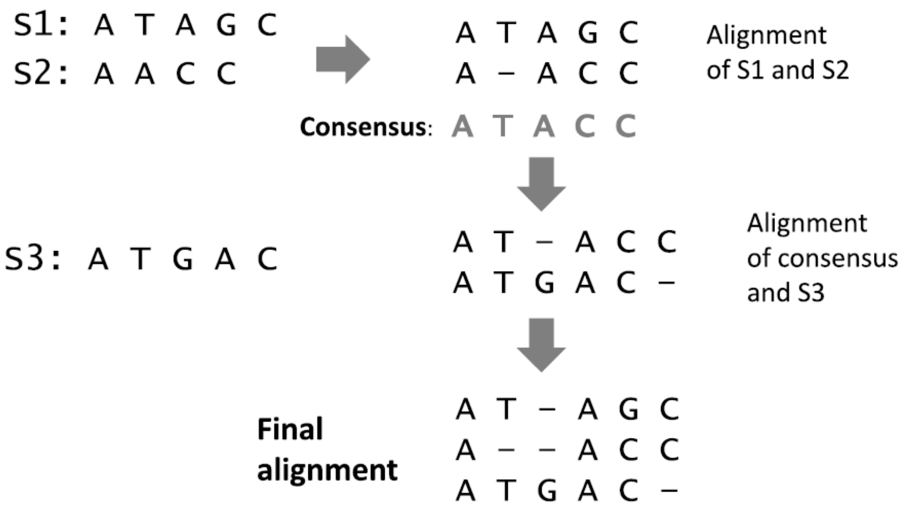
Other Algorithms for MSA

- Ensemble Methods
  - aggregating a large number of different MSA computed by different algorithms for the same set of sequences by focusing on the consensus at each column (e.g. M-COFFEE, MergeAlign)
- Hidden Markov Models
  - Probabilistic models used to represent probabilities of sequential events or sequential data.
  - HMM can be used to model residue distributions in the columns of an MSA and then used to compute the most probable MSA from the given set of sequences.
- Local search methods
  - Optimization algorithms that search for the best solution (e.g. simulated annealing, genetic algorithms)

Implementing Progressive Alignments

Multiple Sequence Alignment: Class MultipleAlign

1. first takes the current alignment and calculates its consensus.
2. then, this consensus is aligned with the new sequence using the Needleman-Wunsch algorithm.
3. the resulting alignment is reconstructed based on the columns of the consensus, filling the column with gaps if the alignment puts a gap in that position.



Summary

- Multiple Sequence Alignment (MSA)
- Scoring Multiple Alignments
- Heuristic Algorithms for Multiple Sequence Alignment
  - Progressive
  - Iterative
  - Others

Further reading

- Ferreira Chapter 8
- Pevzner Chapter 9
- Compeau Chapter 5 (epilogue)